MSBF
Merchant Sales-Based Financing Platform

MERCHANT CREDIT INTELLIGENCE

ENTERPRISE ARCHITECTURE

Enterprise Merchant Sales-Based Financing Platform

From merchant sales to governed funding decisions—and the intelligence to manage what follows.

750
Synthetic merchants
Two matched scenarios • 1,500 application-scenario paths

6,519
Reportable structures
Governed search before a no-feasible conclusion

13.7% / 3.6%
Final-offer authorization rate
Baseline 103/750 • Recession Energy 27/750

15,600
Monitoring account-days
130 funded scenario paths × 120 days

THE ARCHITECTURE • SEVEN FUNCTIONAL BANDS

Primary evidence progression → Bands are not module numbers

0 — Governance & run control (G0 · G1 · all stages)

- G0 — **Physical data foundation**: Schemas, keys, reference policy and typed source structures.
- G1 / M1.1 — **Governed run control**: Run identity, frozen parameters, seed, source and scenario snapshots.
- ALL STAGES — **Validation & acceptance**: Preflight, positive/negative controls, reconciliation and evidence.

1 — Merchant & daily foundation (M1.2–M1.6)

- M1.2–M1.3 — **Merchant + requested terms**: 750 merchants and applications; amount, remittance, payback and horizon.
- M1.4–M1.5 — **POS, settlement & deposits**: Daily sales, cash movements and liquidity; raw evidence stays distinct.
- M1.6 — **Matched scenario panel**: Baseline and Recession Energy preserve merchant and application identity.

2 — Evidence, features & capacity (M1.7–M1.12)

- M1.7–M1.8 — **Quality, verification & continuity**: Separate usable evidence, identity/fraud signals and operating continuity.
- M1.9–M1.10 — **Cash flow & merchant capacity**: As-of features, obligations, residual cash flow and post-financing liquidity.
- M1.11–M1.12 — **Resilience & comparative risk**: Operating archetypes and risk components; transparent synthetic proxy.

3 — Economics & G2 contracts (M1.13–M1.17)

- M1.13–M1.14 — **Exposure, loss & economics**: Path-weighted exposure, recovery, comparative loss, costs and contribution.
- M1.15–M1.16 — **Two consumption contracts**: Scenario-aware application evidence + acquisition attribution/cost companion.
- M1.17 / G2 — **Accepted evidence foundation**: Read-only 1,500-path interface; contracts, archives and source identity assured.

4 — Decision & counter-offer strategy (M2.1–M2.3)

- M2.1 — **Eligibility & policy routing**: Preserve evidence limits and hard stops; route decisionable, remediable paths.
- M2.2 — **Governed structure search**: 6,519 reportable structures; amount, remittance, payback and 30/60/90 days.
- M2.3 — **Final decision & authority**: Seven outcomes. Authorized counter-offers remain distinct from review.

5 — Funded lifecycle & servicing (M2.4–M2.9)

- M2.4–M2.5 — **Fund, activate & monitor**: 130 funded paths; 15,600 account-days. Seasoned pace + early hard triggers.
- M2.6–M2.8 — **Intervene, set up & execute**: Evidence-led review routing, permitted setup and synthetic servicing events.
- M2.9 — **Reconcile & certify accounts**: Events, returns/retries, remaining balances and permitted account states.

6 — Portfolio & certified consumption (M2.10–M2.12 · Power BI)

- M2.10 — **Portfolio performance endpoint**: Source-owned exposure, state, servicing and comparative portfolio KPIs.
- M2.11 — **Bounded terms & account simulation**: Offer/terms and account-treatment comparison across eight profiles.
- M2.12 → P01–P04 — **G3-v2 → Power BI**: 70/70 source controls PASS. Analytical views consume certified source facts.

DELIVERY & STRATEGIC HORIZONS

CURRENT TECHNICAL DELIVERY

- **Module 1 / G2** — Accepted evidence foundation
- **Module 2 / R26** — Accepted technical successor
- **Power BI R3** — P01–P04 complete

PREPARATION & DEVELOPMENT

Campaign Scale: 750 replay → 2,500 → 25,000

Module 3 development • separate acceptance and execution gates

CONCEPTUAL MATURITY HORIZONS

YEAR 1 — Control & learn
Define appetite, test evidence and decisions, observe account behavior.
Build a disciplined foundation.

YEAR 3 — Refine & compare
Extend cohort/vintage depth; compare scenarios, segments and allocation.
Scale only through governed gates.

YEAR 5+ — Enterprise intelligence
Forecasting, scenario planning, relationship insight and governed learning across portfolios.
Broader insight. Better-connected decisions.

FUTURE MODULE CONCEPTS

M4 Platform & Governance • M5 Advanced Intelligence
M6 Strategy Laboratory

Source & parameter identity
Frozen inputs; explicit run lineage.

Positive + negative controls
Expected behavior and safe failure.

Exact reconciliation
Keys, counts, events and balances.

Consumption contracts
Stage-owned facts; explicit boundaries.

GOVERNED CONSUMPTION • EACH HANDOFF RETAINS ITS SOURCE

Module 1 / G2 — Accepted merchant evidence → **M2 / R26** — Decision + synthetic lifecycle → **G3-v2** — Certified source consumption → **Power BI P01–P04** — Executive analytical views → **Future consumers** — Separate contracts / authority

Evidence. Decisions. Accountability.
Independent portfolio • Andrew R. Goad

v2.1 | ENTERPRISE ARCHITECTURE

Deterministic synthetic evidence • Non-production • No real funds